

PONTIAC, IL, USA

WMO: 722171

Lat: 40.924N

Lon: 88.625W

Elev: 201

StdP: 98.93

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04889

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-16.2	-22.6	0.5	-18.2	-19.9	0.7	-15.6	14.5	-2.3	12.8	-2.5	3.1	300	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.6	23.6	32.4	23.3	31.1	22.8	26.5	30.5	25.5	29.7	24.7	28.8	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	21.1	28.8	24.1	19.5	27.9	23.3	18.6	27.0	83.9	30.6	79.3	29.7	75.7	28.6	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	11.0	9.4	-23.9	35.3	4.5	1.6	-27.2	36.4	-29.9	37.4	-32.4	38.3	-35.7	39.5		
(5)			-24.1	28.4	4.3	1.0	-27.2	29.1	-29.7	29.7	-32.1	30.3	-35.2	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-3.3	-2.1	4.9	11.1	17.7	22.5	23.5	22.4	19.6	12.3	5.4	-0.6		
	DBStd	10.98	7.07	6.85	6.31	5.21	5.06	3.30	3.30	3.22	4.02	5.28	5.47	6.08		
	HDD10.0	1512	414	342	184	48	3	0	0	0	0	34	157	329		
	HDD18.3	3177	672	573	419	222	76	4	2	5	32	198	388	586		
	CDD10.0	1947	1	2	27	81	241	376	418	384	288	107	19	2		
	CDD18.3	568	0	0	2	5	56	131	161	131	71	13	0	0		
	CDH23.3	5719	0	0	22	92	598	1273	1665	1207	717	144	1	0		
	CDH26.7	2058	0	0	4	15	218	468	642	414	257	39	0	0		
(14) Wind	WSAvg	4.1	5.2	5.0	5.0	5.4	4.6	3.5	2.4	2.1	2.7	4.1	4.7	4.8		
(15) Precipitation	PrecAvg	878	45	34	65	87	101	97	102	83	84	64	68	56		
	PrecMax	1244	157	87	130	200	255	295	219	220	300	188	208	222		
	PrecMin	537	0	1	15	9	16	7	11	8	1	2	17	8		
	PrecStd	161	32	21	31	41	59	61	51	54	62	43	44	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	18.0	27.0	28.6	33.7	34.8	37.0	34.9	33.6	31.0	22.5	17.0		
		MCWB	13.7	14.7	18.1	17.3	20.9	23.6	25.3	24.0	22.8	21.7	15.1	14.9		
	2%	DB	12.1	13.9	21.3	26.2	31.3	32.8	33.0	32.6	32.1	27.3	18.9	13.0		
		MCWB	10.7	11.4	15.5	16.5	20.0	23.1	25.4	23.9	22.5	18.2	14.6	11.0		
	5%	DB	8.3	9.7	17.8	23.1	29.0	31.4	32.1	31.0	29.9	24.1	16.8	10.1		
		MCWB	7.1	7.7	13.9	15.2	19.2	22.6	24.8	23.4	21.4	17.4	13.3	8.7		
	10%	DB	5.8	6.4	14.9	21.0	27.1	29.1	30.1	28.8	27.7	21.5	13.9	7.7		
		MCWB	4.4	4.6	11.5	14.4	18.6	21.8	24.0	23.0	19.6	16.6	10.8	6.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.5	18.8	19.8	23.6	26.9	28.4	27.2	25.5	22.9	17.8	15.6		
		MCDB	15.6	17.4	24.6	24.9	28.5	30.8	32.1	31.3	30.6	27.7	20.1	16.6		
	2%	WB	10.8	11.6	16.6	18.0	22.4	25.3	27.0	25.8	24.2	21.1	15.2	11.9		
		MCDB	11.9	13.9	20.3	22.8	27.6	29.5	30.9	29.9	29.0	24.7	17.6	12.6		
	5%	WB	7.3	7.7	14.7	16.6	21.4	24.3	25.8	25.0	23.0	19.4	13.4	8.7		
		MCDB	8.1	9.4	17.6	20.8	26.1	28.8	29.9	28.9	26.7	22.2	16.6	9.8		
	10%	WB	4.2	4.8	11.8	15.2	20.2	23.4	25.0	24.1	21.7	17.4	11.3	6.3		
		MCDB	5.5	6.5	14.5	20.6	24.6	27.8	29.2	27.5	25.6	20.4	13.9	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	10.1	12.2	12.1	11.3	11.1	11.4	13.2	11.8	9.7	7.7		
		MCDBR	10.7	13.0	13.8	16.2	14.8	13.0	12.2	13.3	14.8	15.3	13.3	10.6		
		MCWBR	9.3	10.1	9.1	8.4	6.5	5.6	5.6	6.0	6.1	7.8	9.0	9.1		
		MCWBR	10.3	12.1	12.6	13.3	11.9	10.7	10.5	10.4	11.8	11.5	11.7	10.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.293	0.310	0.345	0.395	0.427	0.440	0.455	0.446	0.409	0.355	0.312	0.293		
		taud	2.436	2.388	2.384	2.250	2.219	2.227	2.167	2.181	2.259	2.413	2.485	2.451		
		Ebn at Noon	869	902	905	876	853	840	822	818	824	839	839	837		
		Edh at Noon	78	96	108	132	140	139	146	140	121	93	75	72		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.74	2.63	3.63	4.71	5.39	6.14	6.13	5.40	4.57	3.06	2.01	1.47		
		RadStd	0.19	0.29	0.28	0.38	0.43	0.43	0.33	0.33	0.37	0.34	0.18	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page